

PROJECT DESIGN PHASE – PART 2

Sports Hall Booking System (ServiceNow)

Duration: 3 June 2026 – 5 June 2026

1. Use Case Diagram (Description)

Actors: Student, Sports Coordinator, Administrator.

Student logs in, views halls, books slots, tracks status. Coordinator approves/rejects bookings.

Administrator manages halls, slots, users, and reports.

2. Entity Relationship (ER) Design

Main entities: User, Sports Hall, Time Slot, Booking. A User creates many Bookings. A Sports Hall contains many Time Slots. Each Booking is associated with one Hall and one Time Slot.

3. Database Design

Tables: Sports Hall, Time Slot, Booking. Key fields include Hall Name, Location, Capacity, Slot Date, Start Time, End Time, Booking Status, Requested By, Approval Status.

4. User Interface Design

Interfaces include Login Page, Hall List, Booking Form, Approval Screen, Admin Dashboard, Reports Dashboard.

5. Navigation Flow

Login → Dashboard → View Halls → Select Slot → Booking Form → Approval → Notification → Reports.

6. Module Relationships

Hall Management connects with Time Slot Management. Booking depends on Hall and Slot availability. Flow Designer manages approvals and Notifications inform users.

7. Design Summary

The design provides a modular and scalable architecture that supports efficient sports hall reservation and administration.